

Collatz-Equivalence and a Hardness Ordering for the Collection

A formal treatment: in what precise sense our machines are “Collatz-like,” whether they are equally hard, and an explicit ranking

0. The question, made precise

“Is our machine equivalent to Collatz, and are the machines equally hard?” splits into three formally distinct questions, which the informal “cryptid = Collatz-like” slogan runs together:

(i) Same map?	Is the machine (a conjugate of) the $3n + 1$ map itself?
(ii) Same class?	Is it a member of the class Conway and Kurtz–Simon proved undecidable — the formal content of “Collatz-like”?
(iii) Same hardness?	Is deciding its halting as hard as the Collatz conjecture — in logical complexity, or in proof obstruction?

The short answers, developed below: **(i) No** — none of our machines is the Collatz map. **(ii) Yes** — all are generalized Collatz functions, and that is exactly what makes them Collatz-like. **(iii) Equal at the core, ranked on the shell** — their non-halting statements sit at the same logical level (one quantifier *below* the Collatz conjecture, not at it) and are blocked by the same pseudorandom-orbit obstruction, so on the deepest axis they tie; but a genuine partial order appears on secondary structure, and it is explicit (Section 4).

1. Same class: all are generalized Collatz functions

Definition (Kurtz–Simon [2], after Conway [1]). A *Collatz function* is g with a modulus p and rationals a_i, b_i such that $g(x) = a_i x + b_i$ whenever $x \equiv i \pmod{p}$, always integer-valued. Matthews–Watts [4] give the standard d -branch form $T(x) = (m_i x + r_i)/d$ for $x \equiv i \pmod{d}$. Each is **piecewise-affine with the branch chosen by a residue modulo a fixed number** — finitely many affine pieces.

Every machine in the collection is of this form at the base level: each rule is affine in the state, and the rule is selected by residues (and, in machines 1 and 4, magnitude comparisons) of the current state. So all belong to the class whose halting problem Conway [1] proved undecidable, and whose totality set $\{g : \forall x \exists i \ g^{(i)}(x) = 1\}$ Kurtz–Simon [2] proved Π_2^0 -**complete**. This membership is the formal meaning of “Collatz-like”; the bbchallenge notion of “cryptid” is admittedly informal [6], and the class above is the only precise formalization in the vicinity.

A subtlety our own analysis exposes. The **reduced** return maps (machine 1’s F , the reset maps, the Space Needle’s step) branch not on a fixed residue but on a q -adic **valuation** or a whole **digit word** — unboundedly many affine pieces. Valuation-branched maps are *not* in the strict finite-residue class [confirmed against 2, 4]; they are a broader family (Conway’s functional / FRACTRAN model still contains them). So the acceleration is what *exposes* the machines’ true branch complexity: finite-residue at the base, valuation/digit at the section. The lone exception is the Hydra function $H(n) = \text{floor}(3n/2)$, which branches only on parity — a **strict** Kurtz–Simon Collatz function; Hydra and Antihydra are its two boundary conditions.

(i) **None is the Collatz map.** The maps are pairwise distinct and none is affinely conjugate to $n \rightarrow n/2, 3n+1$. Two internal coincidences are worth recording as genuine equivalences: **Hydra** $\equiv$ **Antihydra** (the same map H , differing only in start and in the direction of the count), and **machine 3** $\approx$ **Space Needle** (the same multiplicative structure, base 3 versus base 2 — machine 3 was built as the analogue). Machines 1 and 4 are distinct singletons.

2. Same hardness? The arithmetical level is not the same

The load-bearing clarification. Fix the start; write the halting question as a logical sentence:

statement	logical form	level	scope
our machine halts from s_0	$\exists k$ [halted by step k]	Σ^0_1	single orbit
our machine never halts	$\forall k$ [not halted by step k]	Π^0_1	single orbit
a single Collatz orbit reaches 1	$\exists i$ [$g^i(n)=1$]	Σ^0_1	single orbit
the Collatz conjecture	$\forall n \exists i$ [$g^i(n)=1$]	Π^0_2	all orbits

So each machine’s non-halting is a Π^0_1 statement — a single universal quantifier over one orbit — exactly as for the Busy Beaver cryptid Antihydra, whose non-halting is $\forall n[\#\text{odd}(n) \leq 2 \cdot \#\text{even}(n)]$ over one iteration of $\text{floor}(3n/2)$ [6], and exactly the form Aaronson–Yedidia [5] use for a specific machine independent of ZFC. The Collatz **conjecture**, by contrast, is Π^0_2 ($\forall n \exists i$), and is *not* known to drop a level: that would require a proven computable bound on stopping times, and none exists (the $\leq 41.68 \log n$ bound is a heuristic prediction of stochastic models, not a theorem [3]).

Consequence. “As hard as Collatz” is *false* read as arithmetical complexity: our machines (and Antihydra) are one quantifier *simpler* than the conjecture. The honest same-level comparison is **single cryptid orbit** $\leftrightarrow$ **single Collatz orbit** (both Σ^0_1/Π^0_1). What “as hard as Collatz” correctly names is the shared **proof obstruction**, not the logical form.

3. Why single-instance equivalence is vacuous — and where the real content is

A single fixed instance (“does Antihydra halt?”, “does Collatz hold?”) has a definite truth value: it is one bit, hence **trivially decidable** by a constant algorithm. Many-one and Turing reductions are only defined between *parametrized families*, never between two fixed bits. This is precisely why every undecidability result is stated over the *class* of Collatz functions and never for $3n + 1$ alone [1, 2], and why Conway’s theorem “does not directly imply” anything about the Collatz conjecture itself [3].

So the formal content of “equivalent to Collatz” is: (a) our machines are members of the same undecidable class (Section 1); (b) that class is Π^0_2 -complete, so as a *family* it is exactly as hard as anything definable at that level [2]; but (c) any *individual* member — ours, Antihydra, or $3n + 1$ — is a single sentence whose difficulty is not a reduction-degree but the **obstruction to proving it**. There is no established many-one ordering among specific Collatz-like instances; the literature offers none [6]. The ordering in Section 4 is therefore ours to propose, and we mark it as such.

4. The shared obstruction, and the ranking

The obstruction is uniform. A proof for any of these machines must resolve a *single* orbit of a mixing, pseudorandom process. This is Lagarias’s diagnosis of Collatz verbatim: the pseudorandomness “supports the conjecture and at the same time deprives us of any obvious mechanism to prove it, since mathematical arguments exploit the existence of structure, rather than its absence” [3]. The 2-adic map is conjugate to the shift (maximum entropy, parities are coin flips), but the integers are a measure-zero subset, so ergodicity decides no single orbit [3]. Our explorations report makes this exact for the collection: the branch word is orbit-determined and grows without bound, and the process generating it is mixing with a positive spectral gap. The calibration is $5x + 1$, where *not one* divergent orbit has been proven [3] — the same wall. **On this primary axis all our machines, Antihydra, and Collatz tie.**

A partial order does appear on secondary structure. Since the core ties, we rank by how much is *provable* around it — how certainly each machine is beyond elementary methods, and how close it sits to a named open problem.

4.1 Axis A — certified vs candidate cryptid

For the **multiplicative** machines (3 and the Space Needle) it is a **theorem** that no congruence can decide halting: the branch is the q-adic valuation, which is independent of $x \bmod M$ for every M coprime to q , so no congruence closure even exists (explorations, Finding 3). They are **certified** irreducible to the elementary decider. For the **sparse-coincidence** machines (1 and 4) the same is only **empirical**: no separating modulus was found ($m \leq 256 / 628$), but none is proven impossible — an elementary decision is not ruled out. So:

$$\text{certified (3, Space Needle)} > \text{candidate (1, 4)},$$

where $>$ reads “more certainly a genuine cryptid.” This is not a claim that the core is harder for the multiplicative machines — it is that we have *proven* they resist the one elementary tool, whereas for the sparse machines that resistance is conjectural.

4.2 Axis B — distance to a named open problem

Some machines’ halting is *analogous* to an established open problem, which imports difficulty of known provenance. **Hydra / Antihydra** and **machine 1’s mantissa** live in the $(3/2)^n$ -equidistribution circle around **Mahler’s Z-number problem** [7]: the bbchallenge wiki states the analogy explicitly (parity of $\text{floor}(c \cdot (3/2)^n)$) — though as an *analogy*, not a proven equivalence, and the equidistribution itself is open [6, 7]. The **multiplicative** machines’ “does the orbit hit an exact power?” is perfect-powers-in-sequences territory (Baker; explorations Finding 6). Machine **4** is freestanding. So Axis B ranks Hydra/Antihydra and machine 1 closest to established hard number theory.

4.3 The ordering, assembled

machine	type / branch	Axis A	Axis B	equiv. class
machine 1	sparse / digit-word	candidate	Mahler (via mantissa)	— (singleton)
machine 3	multiplicative / v_3	certified	perfect powers	$\approx$ Space Needle
machine 4	sparse / magnitude	candidate	freestanding	—

machine	type / branch	Axis A	Axis B	equiv. class
Space Needle	multiplicative / v_2	certified	perfect powers	$\approx$ machine 3
Hydra / Antihydra	walk / parity (strict)	n/a (count, not congruence)	Mahler (analogy)	Hydra $\equiv$ Antihydra

Reading the table. There is no single total order — hardness here is genuinely multi-dimensional, and the primary (logical / obstruction) axis is flat. The defensible statements are: all six are generalized Collatz functions (Section 1); all are Π_1^0 single-orbit, one level below the Π_2^0 conjecture (Section 2); and among them the multiplicative pair {machine 3, Space Needle} is the only one *proven* beyond congruences, while the Hydra pair sits closest to a named open problem. That is the honest content of “are they equally hard, or is there a ranking”: **equal at the core, ordered on the shell.**

5. Answers, in one place

question	answer
Is any machine Collatz (the $3n+1$ map)?	No — distinct maps; two internal equivalence classes (Hydra $\equiv$ Antihydra; machine 3 $\equiv$ Needle).
Are they “Collatz-like” formally?	Yes — all are generalized Collatz functions (Conway / Kurtz–Simon); Hydra is strict, the reduced maps are valuation/digit-branched (broader).
Equally hard as the Collatz conjecture?	Not in logical level: Π_1^0 single-orbit vs the conjecture’s Π_2^0 . Equal only in the proof obstruction.
Equally hard as each other?	At the core yes (uniform single-orbit pseudorandom wall); a partial order on secondary structure (Section 4).
Is there a ranking?	Yes, and explicit: certified (3, Needle) > candidate (1, 4) on Axis A; Hydra/Antihydra, machine 1 nearest a named problem on Axis B. No literature precedent — ours to propose.

References. [1] Conway, Unpredictable iterations, 1972; FRACTRAN, 1987. [2] Kurtz & Simon, The undecidability of the generalized Collatz problem, TAMC 2007 (Def. 1.2; Thm 1.5, Π_2^0 -completeness). [3] Lagarias, The $3x+1$ problem: an overview, arXiv:2111.02635 (pseudorandomness; 2-adic conjugacy; $5x+1$ not one orbit proven). [4] Matthews & Watts, Acta Arith. 43 (1984). [5] Aaronson & Yedidia, A relatively small Turing machine independent of set theory, arXiv:1605.04343 (Π_1 non-halting). [6] bbchallenge wiki, Cryptids / Antihydra (informal cryptid notion; no ranking; Mahler analogy). [7] Mahler, An unsolved problem on the powers of $3/2$, 1968; Flatto–Lagarias–Pollington 1995. Machine-side claims: collatz/formal/classify.py, ranking.py; explorations report (Findings 3, 6). Flags: the Π_1 -vs- Π_2 contrast and the ranking are our syntheses; the Antihydra–Mahler tie is an analogy, not a theorem.